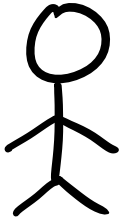


% → per cent

→ per hundred

Shrenik Jain — Study Simplified

#1



[Vinayak]

Total = 100M

% mrvs of V = 10%

Actual mrvs of V = ?

Actual mrvs of V = % mrvs of V x total

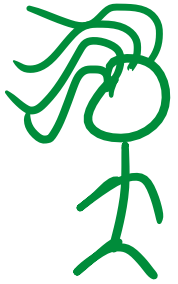
$$= 10\% \times 100M$$

$$= \frac{10}{100} \times 100M = 10M //$$

$$\text{Req. Number} = \text{Req \%} \times \text{total}$$

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2



[Soni]

total = 100M

Soni = 90M

% mrks of Soni = ?

$$\% \text{ mrks of Soni} = \frac{\text{Soni Actual mrks}}{\text{total}} \times 100\%$$

$$= \frac{90}{100} \times 100\% = 90\%$$

$$\% = \frac{Req}{total} \times 100 \%$$

Shrenik Jain — Study Simplified

9] The data given in the following table summarizes the monthly budget of an average household

Category	Amount
Food	4000 ✓
Clothing	1200 ✓
Rent	2000 ✓
Savings	1500 ✗
Others	1800 ✓

The approximate percentage of the monthly budget NOT
spent on savings is

(A) 10%

~~(B) 14%~~

~~(C) 81%~~

(D) 86%

[GATE 2012]

$$\textcircled{1} \quad \% = \frac{\text{Req}}{\text{total}} \times 100 = \frac{9000}{10500} \times 100\% = 85.7\% \leq 86\%$$

M1

$$\begin{aligned} \textcircled{2} \quad \text{Req} &= ₹ (\text{Not spend on saving}) \\ &= 4000 + 1200 + 2000 + 1800 \\ &= 9000 ₹ \end{aligned}$$

$$\textcircled{3} \quad \text{total} = 9000 + 1500 = 10500 ₹$$

% NOT spend on saving

(M2)

$$= 100\% - \% \text{ spend on saving}$$

$$= 100\% - \left[\frac{1500}{10500} \times 100 \right] \%$$

$$= 100\% - 14.28\%$$

$$\approx 85\%$$

In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

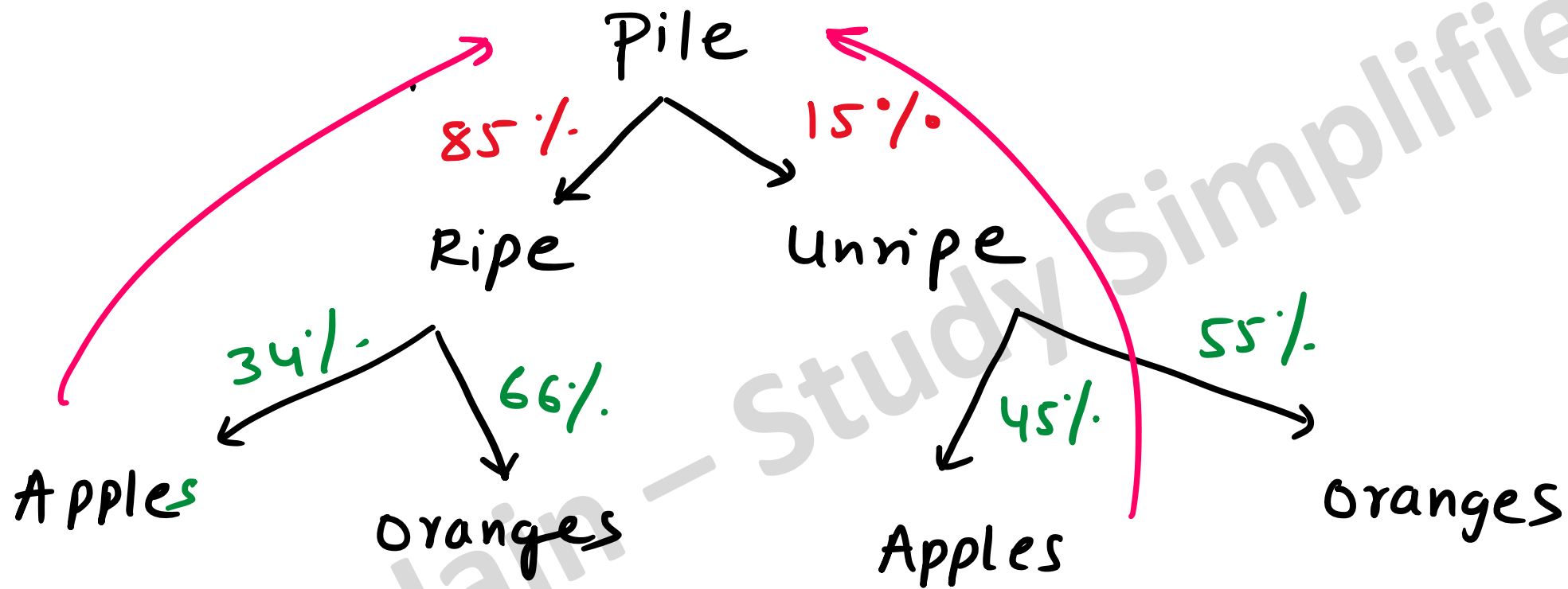
(A) 2029198

(B) 2467482

(C) 2789080

(D) 3577422

[GATE 2016 EC/ME]



$$\text{Apple} = [R\% + UR\%] \cdot \text{total}$$

$$\text{count} = [34\% \text{ of } 85\% \text{ of total}] + [45\% \text{ of } 15\% \text{ of total}]$$

$$\text{Apple} = [R\% + uR\%] \cdot \text{total}$$

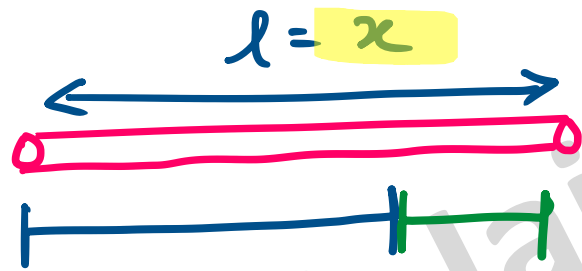
count

$$= \left[\underline{34\% \text{ of } 85\% \text{ of total}} \right] + \left[\underline{45\% \text{ of } 15\% \text{ of total}} \right]$$

$$= \left[\frac{34}{100} \times \frac{85}{100} + \frac{45}{100} \times \frac{15}{100} \right] 5692000$$

2029198

A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece?



$$\text{Shorter piece} = x - 0.7x = 0.3x$$

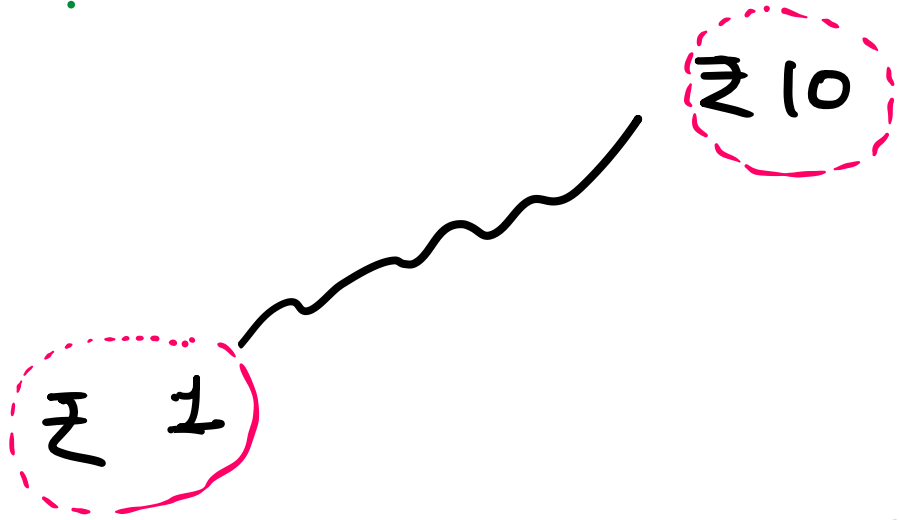
$$\text{Longer piece} = 70\% \text{ of length of pipe}$$

$$= 70\% \text{ of } x = \frac{70}{100} \times x = 0.7x$$

$$\% \text{ change} = \frac{(\text{Longer piece} - \text{Shorter piece})}{\text{Shorter piece}} \times 100\%$$

$$= \frac{0.7x - 0.3x}{0.3x} \times 100\%$$

$$= \frac{400}{3} \%$$



$$\begin{aligned}\% \text{ change} &= \frac{10-1}{1} \times 100\% \\ &= 900\% \quad //\end{aligned}$$

$$\% \text{ Change} = \left| \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}} \right| \times 100$$



$$\% \text{ change} = \frac{1 - 10}{10} \times 100\%$$

$$= -90\%$$
$$= 90\% \text{ [loss]}$$

The income of A is 20% higher than that of B.
The income of B is 25% less than of C. What percent less is A's income from C's income?

Solⁿ:

$$\textcircled{1} \quad \% = \frac{|A - C|}{C} \times 100\%$$

M1

$$\textcircled{2} \quad C's \text{ income} = x$$

$$B = C - 25\% \text{ of } C$$

$$= x - 25\% \text{ of } x = x - 0.25x$$

$$B = C - 25\% \text{ of } C$$

$$= x - 25\% \text{ of } x = x - 0.25x$$

$$\rightarrow \boxed{B = 0.75x}$$

$$\rightarrow A = B + 20\% \text{ of } B$$

$$= 0.75x + \frac{20}{100} \times 0.75x$$

$$\boxed{A = 0.9x}$$

$$\% = \frac{|A - C|}{C} \times 100\%$$

$$= \frac{|0.9x - x|}{x} \times 100\%$$

$$\% \text{ change} = 10\%$$

The income of A is 20% higher than that of B.

The income of B is 25% less than of C. What percent less is A's income from C's income?

(M2)

$$A = B + 20\% \text{ of } B$$

$$= [1 + 20\%] B$$

$$A = \left[1 + \frac{20}{100} \right] B = \frac{6}{5} B$$

$$A = \left[\underbrace{1 + \frac{20}{100}} \right] B$$

Higher 20%.

$$\frac{B}{C} = \left[\underbrace{1 - \frac{25}{100}} \right] C$$

lower 25%.

$$B = C - 25\% \text{ of } C$$

$$= [1 - 25\%] C$$

$$B = \left[1 - \frac{\cancel{25}^1}{\cancel{100}} \right] C$$

$$B = \frac{3}{4} C$$

$$A = \frac{6}{5} B \checkmark$$

$$\rightarrow A = 3 \frac{6}{5} \left(\frac{3}{4} C \right)$$

$$B = \frac{3}{4} C$$

$$A = \frac{9}{10} C$$

$$A = \left[\underbrace{1 - \frac{10}{100}}_{\text{lower 10\%}} \right] C$$

lower 10%

A gardener increased the area of his rectangular garden by increasing its length by 40% and decreasing its width by 20%. The area of the new garden

(A) Has increased by 20%

~~(B) Has increased by 12%~~

(C) Has increased by 8%

(D) Is exactly same as the old area.

$$l' = \left[1 + \frac{40}{100} \right] l$$

$$b' = \left[1 - \frac{20}{100} \right] b$$

$$\textcircled{1} \quad A (\text{old}) = l \times b$$

$$\textcircled{2} \quad A' (\text{New}) = l' \times b'$$

$$A' = \left(\frac{7}{5} l \right) \left(\frac{4}{5} b \right)$$

$$A' = \left(\frac{7}{5} \ 1 \right) \left(\frac{4}{5} \ b \right)$$

$$= \left(\frac{7}{5} \cdot \frac{4}{5} \right) (1 \ b)$$

$$A' = \frac{28}{25} \cdot A$$

$$= [1.12] A$$

$$A' = [1 + 0.12] A$$

$$A' = [1 + 0.12] A$$

$$A' = \left[1 + \frac{12}{100} \right] A$$

Higher 12%

(x % of y) + (y % of x) is equivalent to

$$= \frac{xy}{100} + \frac{xy}{100} = 2 \frac{xy}{100}$$

☒ (A) 2% of xy

(B) 2% of (xy/100)

(C) xy% of 100

(D) 100% of xy

[GATE 2016 CE]

$$= \frac{2}{100} \cdot xy$$

$$= 2\% \text{ of } xy //$$

$$x\% \text{ of } y = \frac{x}{100} \times y = \frac{xy}{100}$$

$$y\% \text{ of } x = \frac{y}{100} \times x = \frac{xy}{100}$$

The ratio of the number of boys and girls who participated in an examination is 4:3. The total percentage of candidates who passed the examination is 80 and the percentage of girls who passed is 90. The percentage of boys who passed is _____

(A) 90

(B) 80.5

(C) 55.5

(D) 72.5

[GATE EE]

$$\textcircled{1} \quad \frac{B}{G} = \frac{4K}{3K}$$

$$\rightarrow B = 4K, \quad G = 3K$$

$$\textcircled{2} \quad \text{Total candidate} = 4K + 3K = 7K$$

$$\% \text{ pass candidate} = 80\%$$

$$\underline{\text{pass candidate}} = \text{pass candidate \%} \times \text{total}$$

$$= 80\% \times 7k$$

$$= \frac{80}{100} \times 7k$$

$$= 5.6k //$$

$$\cdot \text{Girls pass} = \cdot\% \text{ Girls pass} \times \text{total}$$

$$= 90\% \times 3K$$

$$= \frac{90}{100} \times 3K$$

$$= 2.7K$$

$$\checkmark \text{ Boys pass} = 5.6K - 2.7K$$

$$= 2.9K$$

$$\% \text{ Boy pass} = \frac{\text{Req}^{\checkmark}}{\text{total}} \times 100\%$$

$$= \frac{2.9K}{4K} \times 100\%$$

$$= 72.5\% //$$

The radius as well as the height of a **circular cone** increases by 10%. The percentage increase in its volume is _____

(A) 17.1

~~(B) 21.0~~

(C) 33.1

~~(D) 72.8~~

[GATE IN/PI]

(M)

$$r' = \left(1 + \frac{10}{100}\right)r$$

$$r' = 1.1r$$

$$h' = 1.1h$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V' = \frac{1}{3} \pi (r')^2 \cdot (h')$$

$$= \frac{\pi}{3} (1.1r)^2 (1.1h)$$

$$V' = \frac{\pi}{3} (1.1)^3 \cdot r^2 \cdot h$$

$$V' = \left(\frac{\pi}{3}\right) (1.1)^3 \cdot (r^2 \cdot h)$$

$$\Rightarrow V' = (1.331) V$$

$$= [1 + 0.331] V$$

$$V' = \left[1 + \frac{33.1}{100} \right] V$$

Increase 33.1%

∴ (M2)

$$\% \text{ change} = \frac{FV - IV}{IV} \times 100\%$$

$$= \frac{(1.331V) - V}{V} \times 100\%$$

$$= 33.1\% //$$

$$V' = (1.331)V$$

Successive % change

Q] ₹ 300 was increased by 20%. then
it was decreased by 10%. Value now ?

$$\textcircled{1} \quad \text{New value} = \left(1 + \frac{20}{100}\right) 300$$
$$= 360$$

MI

$$\text{Updated New value} = \left(1 - \frac{10}{100}\right) 360$$
$$= 324 //$$

Successive % change

Q] ₹ 300 was increased by 20% then
it was decreased by 10%. Value now?

Solⁿ: New value = old value $\times$ [multiplying factor]

$$= 300 \times \left[1 + \frac{20}{100}\right] \times \left[1 - \frac{10}{100}\right]$$

$$= 324 //$$

M2